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The Reporting of Computation-Based Results in Statistics

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Abstract

Current practices in reporting computation-based results in statistics, especially in statistical theory, can be strengthened substantially. We sketch the role of computers in statistics, discuss the reporting of scientific experiments, examine the current state of reporting in statistics, and propose standards for computation-based results.

In recent years the general availability of powerful digital computers has had widespread impact on statistics. Best known, perhaps, are advances in data analysis: complicated procedures which not so long ago could only be formulated are now frequently, even routinely, applied to large bodies of data. Multivariate analysis is a source of many examples, among them log-linear analyses of multidimensional contingency tables [2, 12] and spectrum analysis, supported by the "fast Fourier transform" [16]. Flexible linear fitting, including regression analysis, has become much more common [5], and nonlinear fitting has become feasible. But computers have contributed far more than brute numerical force—a number of interactive systems for data analysis are in daily use, and developments in interactive graphics (for example, PRIM-9 [9, 23]) are opening up a host of new possibilities for display and analysis of multidimensional data [24]. The processes of computing have themselves become a source of many types of data, often inviting new approaches for their analysis [10, 14].

Less visibly, computers have become an almost indispensable resource for statistical theory, subduing previously intractable calculations. One substantial example arises in using linear combinations of order statistics to estimate location and scale parameters in a specified family of distributions [6, p. 102]. The coefficients in the minimum-variance unbiased linear estimators may be calculated from the Gauss-Markov theorem once one knows the expected values and covariances of the order statistics. These in turn generally present a difficult problem and must be found by numerical integration [11]. The problem of multiple comparisons provides a further example, illustrative of many others. To study the effectiveness or robustness of various techniques, simulation is required [3].

Unfortunately, statisticians have often been far less energetic in documenting and reporting computational results than in pursuing such applications of computers. The situation in statistical theory seems particularly acute. In what follows we first sketch a number of the ways in which computers play a supporting role in statistical research. Then we examine prevailing practices in reporting computation-based results. Finally, we offer some suggestions for strengthening the standards of publication.

Computing and Statistical Theory

Among the many ways of using computers in statistical research, simulation and Monte Carlo are probably the most widespread. Examples may be as simple as randomly generating a few test cases to enhance insight into a new procedure or as complicated as carefully designed comparative studies with sophisticated variance-reduction techniques. These are hardly new, as bibliographies on random-number generation [22] and Monte Carlo [13] show, but they are continually being applied to a widening range of problems.

Deterministic numerical techniques are equally important. Expected values and probabilities pervade statistics, and it is hardly surprising that many of them must be evaluated numerically by various quadrature techniques. Often expressions involve special functions, for which computer approximations exist or must be developed. Maximum-likelihood calculations and other optimization problems require rootfinders and sophisticated function-minimization algorithms. While numerical linear algebra may be somewhat less prominent in theoretical calculations than in data analysis, eigenproblems and simultaneous linear equations frequently arise. Not all computer applications are numerical,

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however: systems for symbolic differentiation and formal integration have become quite effective [20], symbolic manipulation of algebraic expressions [21, chapter VII] can be vital in studying complicated asymptotic expansions, and graphic displays are valuable in studying likelihood surfaces.

In the development of new techniques for data analysis, the capabilities of computers have substantially broadened the range of feasible techniques. We can now experiment with iterative techniques such as jackknifing an already complicated procedure or exploring Bayesian analyses with a variety of sampling and prior distributions.

A number of the above components come together in the relatively simple problem of studying robust estimators of symmetric location [1]. Numerical integration arises in calculating asymptotic variances, while Monte Carlo forms the basis for estimates of small-sample variances and quantiles. Investigation of the behavior of some estimators involves finding roots of non-algebraic equations. Many estimators, for example Cauchy maximum-likelihood, virtually require a computer for their evaluation in all but the simplest cases.

Few readers are likely to be surprised at this broad range of uses for computers in statistical theory, but the poor state of documentation in reporting the results may be startling.

Experiments and Reporting . . .

The requirements for reporting results in statistics based on substantial computation bear a clear resemblance to those faced by other sciences. We pursue this parallel by examining the components of a model for careful reports on scientific experiments.

Experimental setting. This should be described in some detail. Accompanying diagrams should show how apparatus was arranged or where samples or measurements were taken. There should be detail drawings for specially constructed pieces of equipment. Sources and characteristics of any special materials should be recorded. Populations of experimental subjects or laboratory animals must be clearly identified and fully described. The experimenter must explain in detail how experimental and non-experimental factors were controlled. If randomization or blindness is involved (as often in the medical or social sciences), the report must state precisely how it was achieved.

Measuring equipment. The accuracy of any measuring instrument should be fully adequate for the needs of the experiment. It is generally desirable to report the manufacturer and model number as well as any modifications which have been made. If appropriate, the report should give details of calibration and data on measurement accuracy.

Analysis of results. These matters demand very careful attention. Generally the results bear on a theory, a conjecture, or previous experimental evidence, and the report must examine the agreement between them, giving due emphasis to both general and particular behavior. Replication may be an important element, and the experimenter should report standard errors for numerical values whenever appropriate.

We have left this model rather general, and specific guidelines must be interpreted in the context of the particular experiment and its subject matter. For a variety of generally satisfactory examples one can browse through the experimental literature in such journals as *American Journal of Botany, Analytical Chemistry, Applied Physics Letters, Cryobiology, Ecological Monographs, Journal of Applied Physics, Science,* and *Virology.* The basic spirit of this model should be clear: all reporting must make it easy for the reader to assess the quality of the experimental work and the accuracy of the results.

Of course, there is much room for improvement. Details vital for the independent replication which good experimental science requires are sometimes omitted. Techniques of data analysis should be assessed for robustness. As Youden [25] and Joiner [17] have pointed out, more attention should be given to assessment of systematic errors, which may dwarf reported estimates of variability based only on simple replication. Emphasis should be shifted away from reporting only p-values and toward estimating effect sizes with appropriate confidence intervals. All too often the reader comes no closer to the data than a few p-values or regression coefficients. But data should not be withheld from the open court of science. The present authors do not believe that readers should be expected to accept results, no matter how significant, unless the data are published or readily available for further exploration.

Such a stringent standard may raise problems for authors and editors alike, and solving them will require cooperation. When the body of data is not too extensive, it will be most readily accessible if published in or with the paper. The use of microform (such as microfiche or microprint) may relieve some of the pressure on journal space; such techniques are now being used experimentally in chemistry, and their possibilities in statistics should be explored. Larger bodies of data might be deposited in a suitable archive, analogous to the Unpublished Mathematical Tables File maintained by the journal *Mathematics of Computation.* The archive is an attractive alternative because it ensures that the data will continue to be available, centralizes it at a recognized source, and transfers the burden of preserving it from the original experimenter to those who are interested in having access to it. Of course, a reliable archive is not a complete solution: data must not be stored in a way which might compromise the confidentiality of information, and not all data is trustworthy enough to merit preservation. Etzioni [8] and Macdonald [18] have focused on some of these problems, and the discussion can be expected to continue.

. . . in Statistical Theory

Nor can we be complacent about what often in the statistics literature passes for experimentation and reporting. Statisticians (who, of all people, should know better) often pay too little attention to their own principles of design, and they compound the error by rarely analyzing the results of experiments in statistical theory. The computer software used is frequently obsolete, inefficient, or even inaccurate. Random-number generators having known serious defects (for example, RANDU [15, 19, 4], whose output triples (U_n, U_{n+1}, U_{n+2}) have $U_{n+2} - 6U_{n+1} + 9U_n$ equal to an integer from -5 to $+9$ and thus are restricted to lie on only 15 planes in the unit cube, as shown graphically in [24]) reappear with amazing persistence, and reports lamely assert that the results should be adequate for the purpose. Usually the report even fails to mention what computer was used, even though this information can be important in judging the accuracy and reproducibility of results.

These lapses might be less serious if papers containing substantial computational results were rare, or if simulation were seldom practiced. But this is not the case. Indeed, simulation plays much the same role in statistics as experimentation plays in other sciences. A brief look at some data for 1973 makes this clear. The Exhibit gives the results of a survey of papers appearing in the *Journal of the American Statistical Association* and *Biometrika*. Papers in *JASA* Theory and Methods and in *Biometrika* having only straightforward illustrative examples have not been counted among those with numerical results; but in *JASA* Applications, analyses of important bodies of data are often the focus of discussion and have been counted. Roughly half of all papers published in *JASA* Theory and Methods and *Biometrika* in 1973 do incorporate numerical results, and of these over a third make some use of simulation results. The much greater frequency (nearly 90%) of numerical results in the Applications Section of *JASA* is explained by its attention to economic and social data; the ratio of papers with simulation results to total papers is about one to five, roughly the same as in the other two cases. This gives some crude indication of the size of the problem of setting standards for publication of computational results. Of course, we do not suggest that all the papers surveyed are deficient. Some, indeed, illustrate careful design and accurate reporting. (A 1973 paper by Yuen and Dixon [26] is a good example. Their treatment of random-number generation, Monte Carlo techniques, accuracy, and analysis of the results is praiseworthy.) Our concern is simply that such papers are the exception rather than the rule. Indeed, in our survey we found surprisingly few which meet the standards we propose.

EXHIBIT

PREVALENCE OF NUMERICAL AND SIMULATION RESULTS, 1973

	Total Papers	Papers with Numerical Results	Papers with Simulation Results
JASA Applications	55	47	10
JASA Theory and Methods	115	60	22
Biometrika	90	44	20

(Note: Simulation papers are a subset of numerical papers.)

Some Proposed Standards

To take a constructive approach, we propose some basic standards which should strengthen experimentation and reporting in the statistical literature. Our recommendations cover three aspects of the process: computational practices, published reporting, and supporting documentation.

Computational Practices. It is easy to agree that computational practices should reflect the current state of the art in statistical computing, but all too often there are gaps, sometimes serious ones. Some of these may be due primarily to the usual lag in producing high-quality software which incorporates the latest techniques, while others seem to be the result of oversight. In closing such gaps we should try to avoid redeveloping good software which is already available. The task of reporting becomes much easier when the software used is in the public domain, preferably in the form of well-known subroutines, libraries, packages, or systems. This is particularly true for random-number generators: the one who uses the generator should bear the burdens of describing the generator precisely and of establishing its theoretical and empirical properties if this information is not already in the published literature.

Published Reporting. A published report of computation-based results must make it easy for the reader to make reasonable assessments of the numerical quality of the results. Specifically it should include information on

- the accuracy of the results (for example, systematic errors, error bounds, and estimated standard errors),
- the extent of agreement with theoretical results (including, where possible, an analysis of departures),
- the details of any simulation (such as uniform pseudo-random-number generators, transformations to non-uniform variates, and any Monte Carlo "swindles"),
- the numerical algorithms used (for such calculations as rootfinding, minimization, quadrature, and linear algebra), and
- the computer(s), programming language(s), and major software components used.

Other information may be appropriate in specific cases, and authors should not hesitate to volunteer it. It is the author's responsibility to convince the reader that the results are correct.

Supporting Documentation. When computation-based results are presented, a conscientious referee has a particularly demanding task. In addition to his usual duties he must form some hard judgments about the quality of the numerical work. Authors should routinely submit supporting documentation to ease this burden. (Incidentally, readers might often find it valuable to know that, as a matter of policy, the referees had carefully examined the computations and results.) In some cases part of the information suggested above may be too extensive to be included in the paper; it then should naturally become supporting evidence for the referees. Generally authors should be prepared to provide such materials as

- detailed studies of accuracy or errors,
- program listings, and
- output from a typical computer run;

and referees should not hesitate to request this information.

Whenever possible, this supporting documentation should be available on request from the author after the paper has been published. Ideally, we feel this should be a condition to publication. Practically, making such documentation easily available to editors and referees would be a major step forward. In some cases a few exceptionally valuable parts of the supporting material might accompany the paper if microform were being used to publish details. Otherwise this documentation might be accommodated in an archive; it might even be required to establish the trustworthiness of the "data".

Some aspects of these proposed standards deserve further discussion. By and large we are suggesting changes which could consume more than a negligible amount of space in print—at a time when many journals face rising costs. We feel, however, that full and accurate reporting is as essential here as in other sciences and that papers based on calculations not worthy of this careful treatment are not worthy of publication in a journal with high standards.

We recognize that it is often difficult to give error bounds, but this information is valuable and should be included when possible.

The position replication occupies in statistical experimentation is not so different from that in other sciences, where three or more independent verifications might be required to confirm a result. Statistical experiments may often have some of the kinds of systematic errors Youden described [25], but usually no professional credit is available for "confirming" someone else's results. This makes it all the more important to vary components (for example, quadrature routines and random-number generators) of the computation, to compare the results with established theory and previously computed values, and to report uncertainties as suggested by Eisenhart [7]. It can also place the referee in an even more central position. Ways of assuring the conscientious referee appropriate recognition deserve serious discussion.

Our suggestions for supporting documentation might cause problems for authors who use proprietary software and thus may be unable to distribute program listings. (This is one of the reasons why we recommend using software in the public domain.) It may still be possible, however, for referees to examine such material without sacrificing its proprietary nature. If this documentation proved necessary to the refereeing process, there could be a standard mechanism by which the journal and the referee agree with the software's proprietor not to copy or divulge any proprietary components or use them outside the refereeing process. In other instances the referee might ask the author to run particular test cases.

An Example

Now to show how these standards might be applied, we sketch an example—a study on small-sample variances of location estimators. In this hypothetical (but currently very common) example we are concerned with the comparative behavior of a moderate number of estimators, some of which are related by being members of one-parameter families. Being careful to include for calibration some well-chosen cases in which the results are independently known, we study their variances for several small sample sizes and several underlying distributions. We naturally describe all these in the report, but since we have engaged in a simulation study, we must provide further details.

Thus we use a reasonable uniform random-number generator from a published source, and we describe the techniques used to obtain the distributions we are studying. We are fortunately able to exploit certain features of the problem and develop effective Monte Carlo techniques, so we describe them, reporting also the number of replications in each case and the efficiency of the Monte Carlo techniques in variance reduction.

As our basic results we report the estimated small-sample variances of the estimators and also their estimated standard errors. We then analyze this data according to the factors in our "design"—estimator, distribution, and sample size, and also the parameter value within the one-parameter families of estimators.

As supporting materials we submit copies of the computer programs for the individual estimators and the basic Monte Carlo techniques. We also document and preserve these so that we may later send copies to interested readers on request.

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These proposed standards and practices have evolved from our experience and that of others. We earnestly hope that this discussion will continue and will contribute to advancing the state of the art, both in statistical publication and in statistical computing.

REFERENCES

[1] D. F. Andrews, P. J. Bickel, F. R. Hampel, P. J. Huber, W. H. Rogers, and J. W. Tukey: *Robust Estimates of Location: Survey and Advances.* Princeton: Princeton University Press, 1972.

[2] Yvonne M. M. Bishop, Stephen E. Fienberg, and Paul W. Holland: *Discrete Multivariate Analysis: Theory and Practice.* Cambridge: MIT Press, 1975.

[3] S. G. Carmer and M. R. Swanson: "An Evaluation of Ten Pairwise Multiple Comparison Procedures by Monte Carlo Methods," *Journal of the American Statistical Association 68* (March 1973), 66–74.

[4] Robert L. Coldwell: "Correlational Defects in the Standard IBM 360 Random Number Generator and the Classical Ideal Gas Correlation Function," *Journal of Computational Physics 14,* 2 (February 1974), 223–226.

[5] Cuthbert Daniel and Fred S. Wood: *Fitting Equations to Data.* New York: Wiley-Interscience, 1971.

[6] H. A. David: *Order Statistics.* New York: John Wiley and Sons, 1970.

[7] Churchill Eisenhart: "Expression of the Uncertainties of Final Results," *Science 160* (14 June 1968), 1201–1204.

[8] Amitai Etzioni: "Who Should Ultimately Own the Data?" *Science 176* (14 April 1972), 121.

[9] Mary Anne Fisherkeller, Jerome H. Friedman, and John W. Tukey: "PRIM-9, An Interactive Multidimensional Data Display and Analysis System," AEC Scientific Computer Information Exchange Meeting, May 2–3, 1974.

[10] Walter Freiberger (Editor): *Statistical Computer Performance Evaluation.* New York: Academic Press, 1972.

[11] J. L. Gastwirth and M. L. Cohen: "Small Sample Behavior of Some Robust Linear Estimators of Location," *Journal of the American Statistical Association 65* (June 1970), 946–973.

[12] Shelby J. Haberman: *The Analysis of Frequency Data.* Chicago: University of Chicago Press, 1974.

[13] J. H. Halton: "A Retrospective and Prospective Survey of the Monte Carlo Method," *SIAM Review 12,* 1(January 1970), 1–63.

[14] David C. Hoaglin: "An Analysis of the Loop Optimization Scores in Knuth's 'Empirical Study of FORTRAN Programs'," *Software—Practice and Experience 3* (April–June 1973), 161–169.

[15] IBM Corporation: *System/360 Scientific Subroutine Package* (360A-CM-03X) *Version III, Programmer's Manual.* H20-0205-3, 1968.

[16] Gwilym M. Jenkins and Donald G. Watts: *Spectral Analysis and Its Applications.* San Francisco: Holden-Day, 1968.

[17] Brian L. Joiner: "Critical Evaluation of Numerical Data," *The American Statistician 28,* 1(February 1974), 14–16.

[18] J. Ross Macdonald: "Are the Data Worth Owning?" *Science 176* (30 June 1972), 1377.

[19] George Marsaglia: "The Structure of Linear Congruential Sequences," in S. K. Zaremba, ed., *Applications of Number Theory to Numerical Analysis.* New York: Academic Press, 1972.

[20] Joel Moses: "Symbolic Integration: The Stormy Decade," *Communications of the ACM 14,* 8(August 1971), 548–560.

[21] Jean E. Sammet: *Programming Languages: History and Fundamentals.* Englewood Cliffs, N.J.: Prentice-Hall, 1969.

[22] E. R. Sowey: "A Chronological and Classified Bibliography on Random Number Generation and Testing," *International Statistical Review 40,* 3(December 1972), 355–371.

[23] S. Steppel (Editor): "PRIM-9," film produced by Bin 88 Productions. Stanford, California: Stanford Linear Accelerator Center, April 1973.

[24] Roy E. Welsch: "Graphics for Data Analysis," to appear in *Computers and Graphics,* 1975.

[25] W. J. Youden: "Enduring Values," *Technometrics 14* (February 1972), 1–11.

[26] Karen K. Yuen and W. J. Dixon: "The Approximate Behavior and Performance of the Two-sample Trimmed t," *Biometrika 60,* 2(August 1973), 369–375.

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The American Statistician attempts to publish articles of general interest to the statistical profession. These articles should be timely and of sufficient generality to appeal to large numbers of readers. Articles should ordinarily be of an expository, not highly technical nature. The following list illustrates topics that would be appropriate:

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